

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

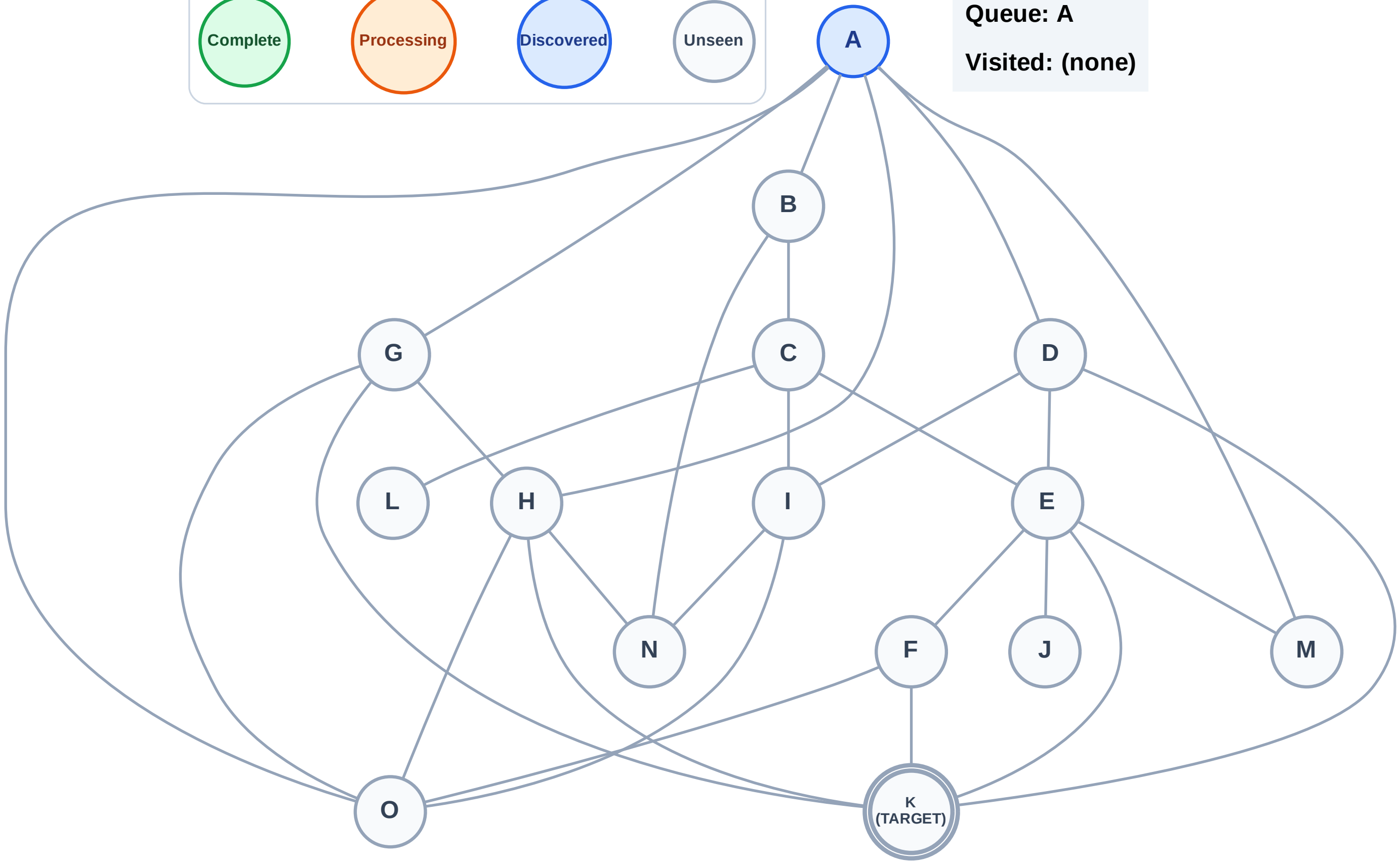
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

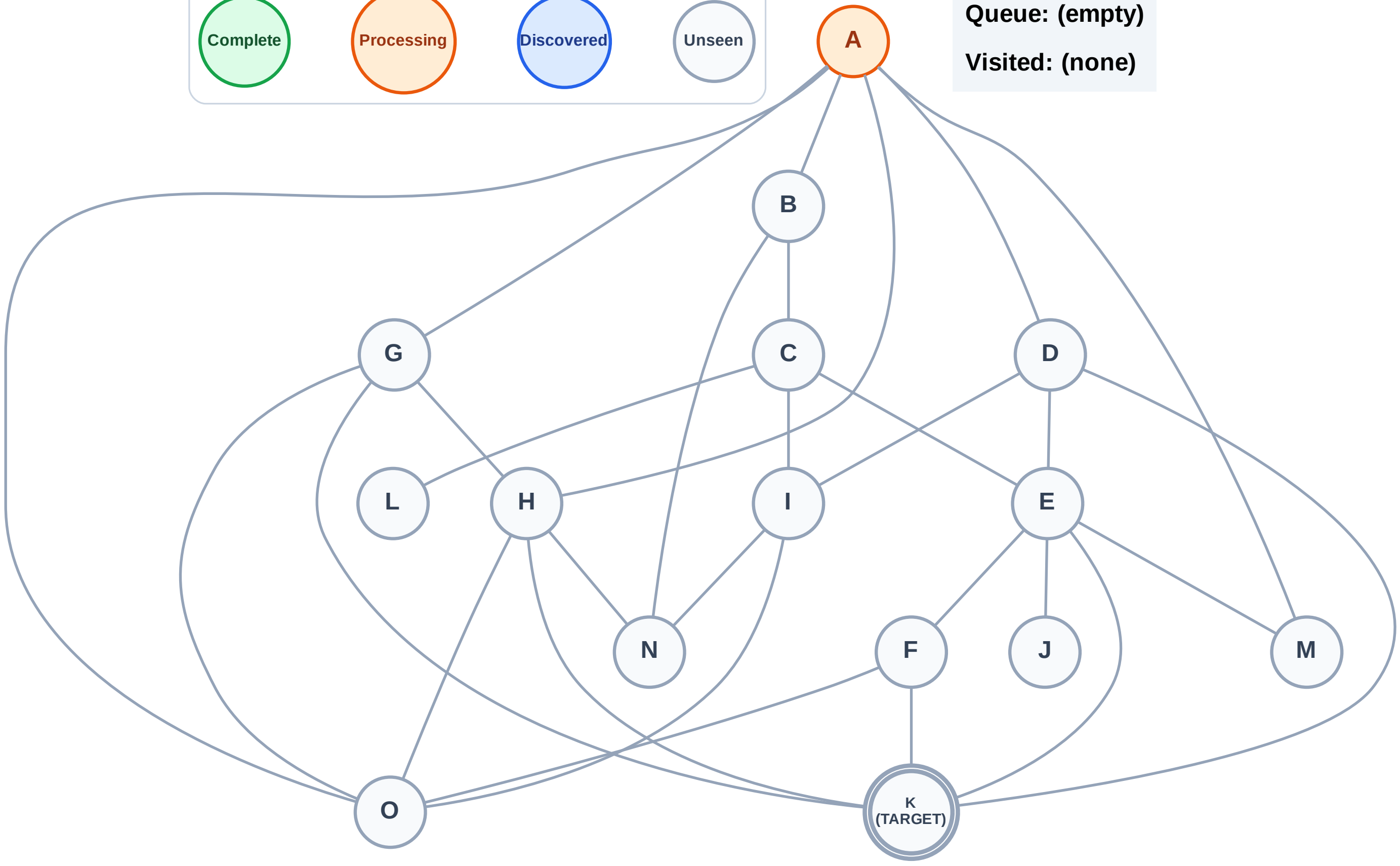
Complete

Processing

Discovered

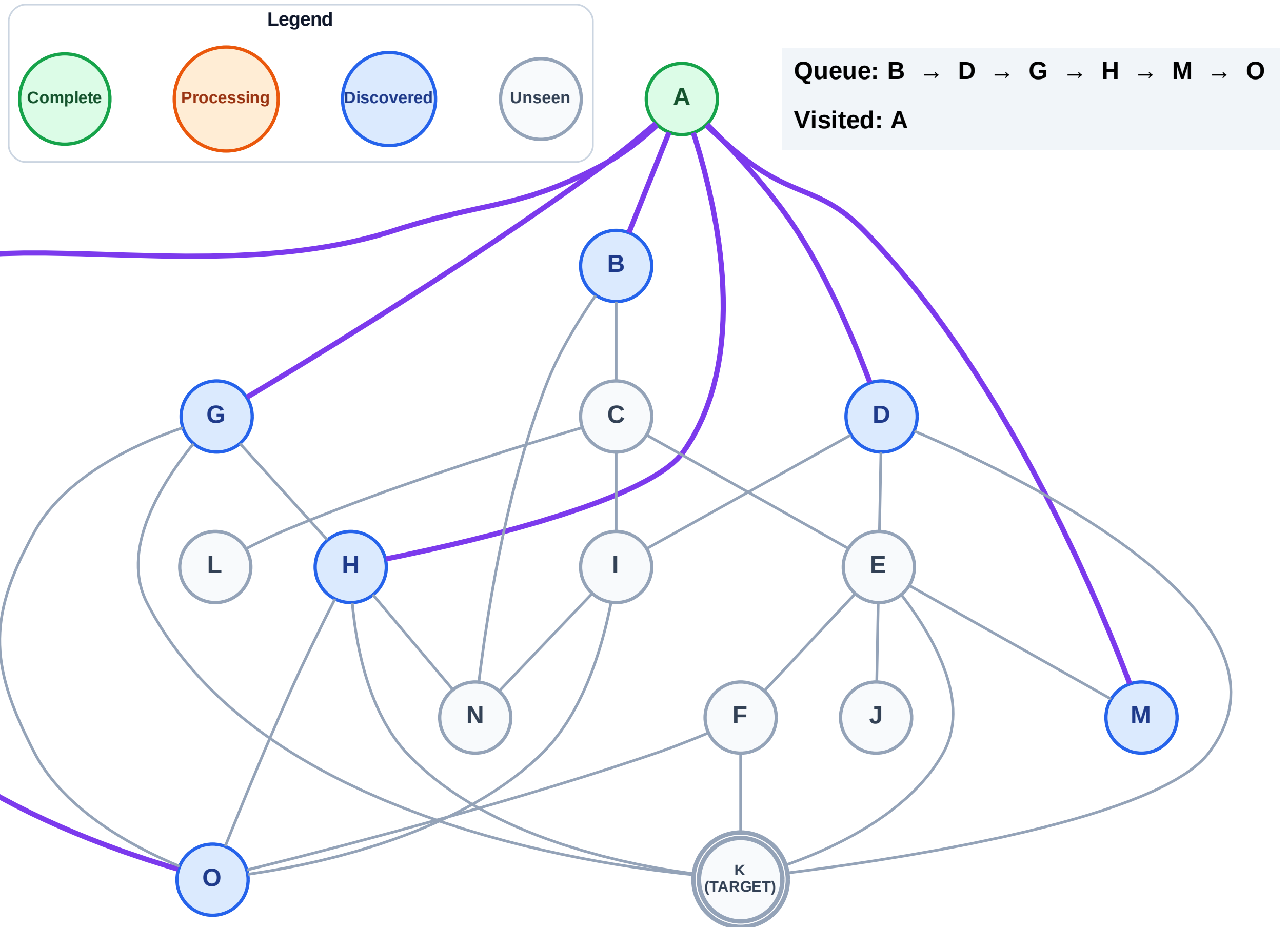
Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, G, H, M, O from A



BFS Traversal

Step 4: Process node B

Legend

Complete

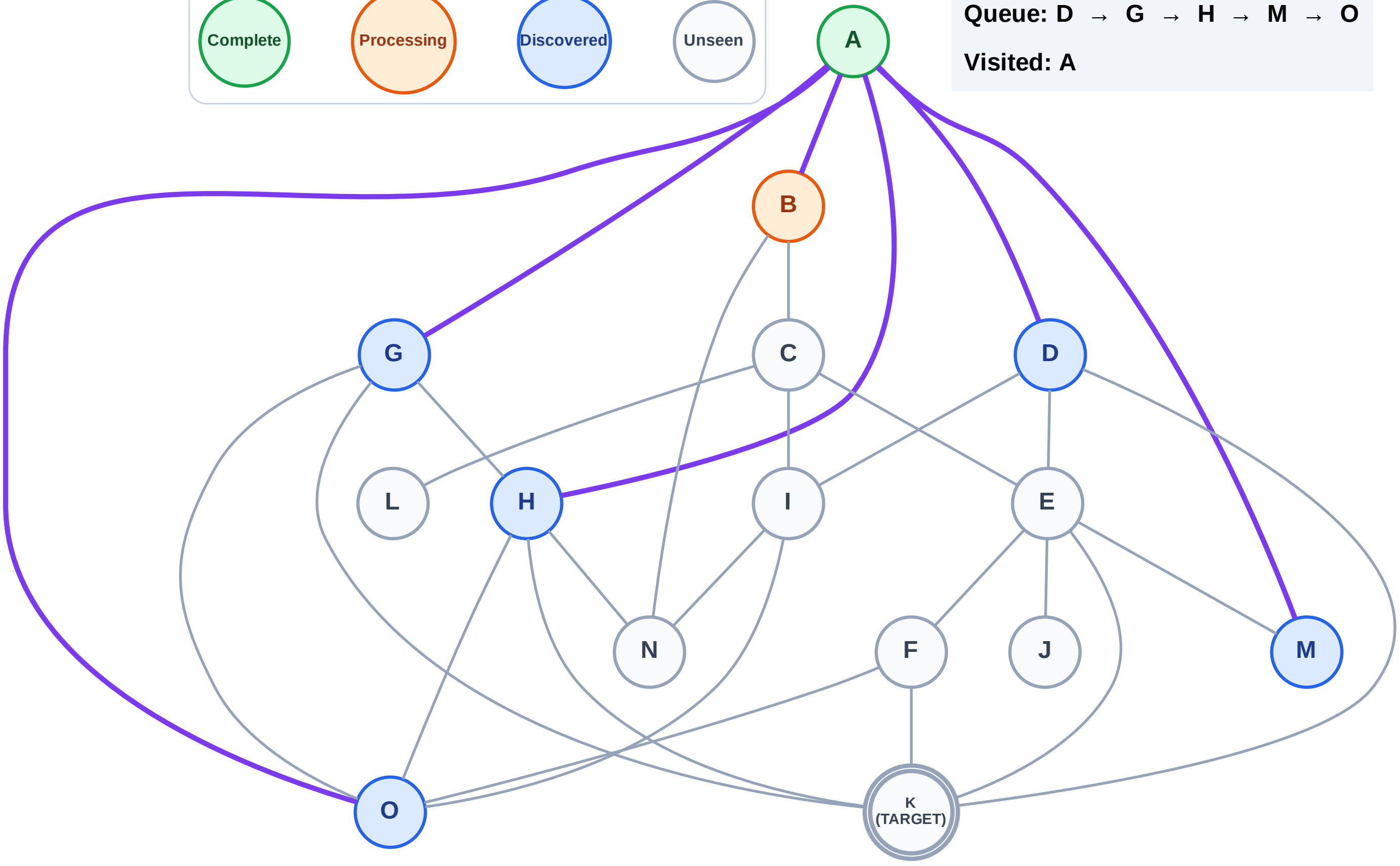
Processing

Discovered

Unseen

Queue: D → G → H → M → O

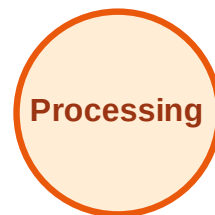
Visited: A



BFS Traversal

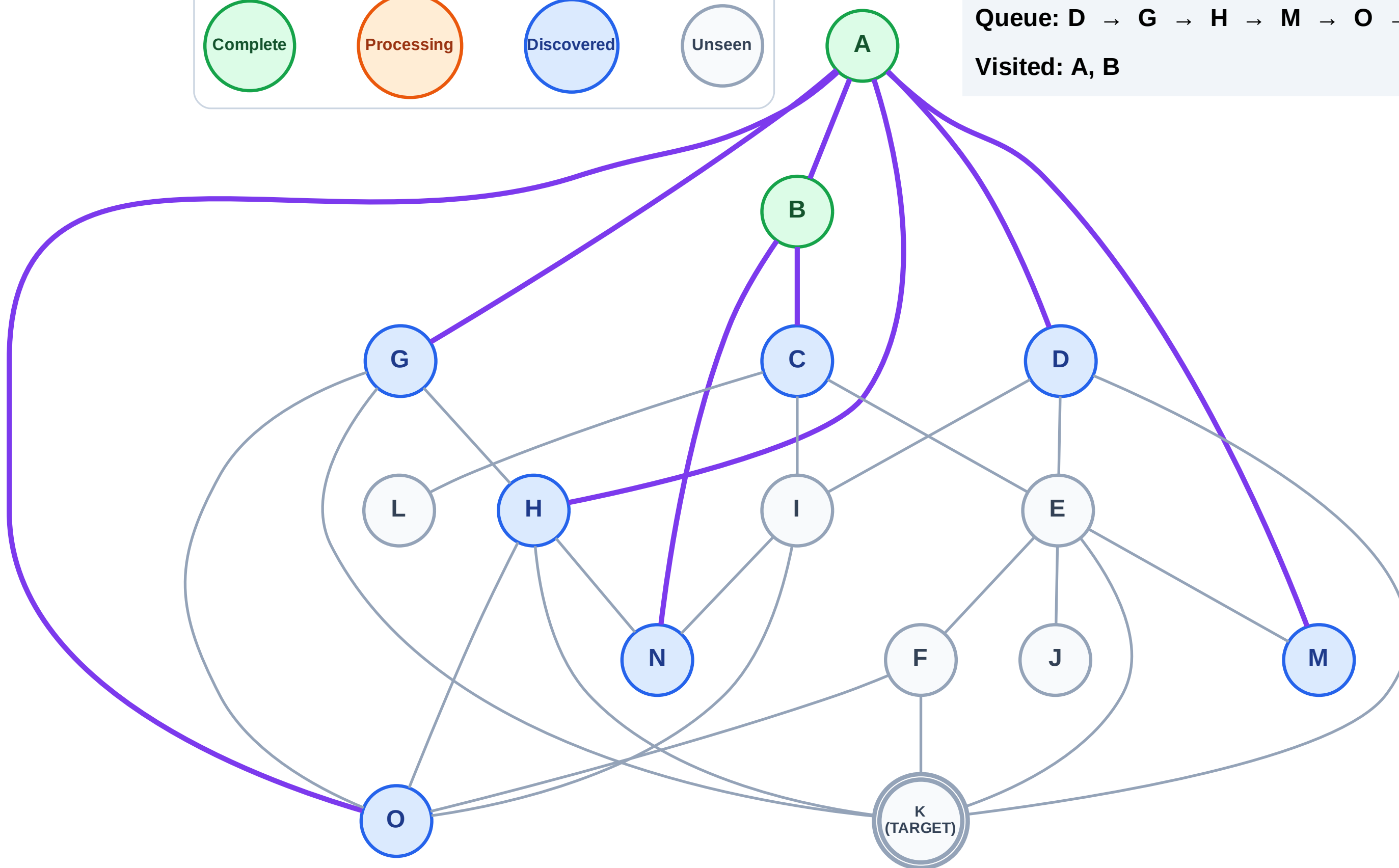
Step 5: Discover C, N from B

Legend



Queue: D → G → H → M → O → C → N

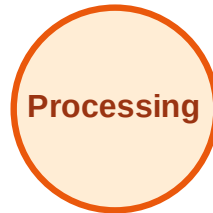
Visited: A, B



BFS Traversal

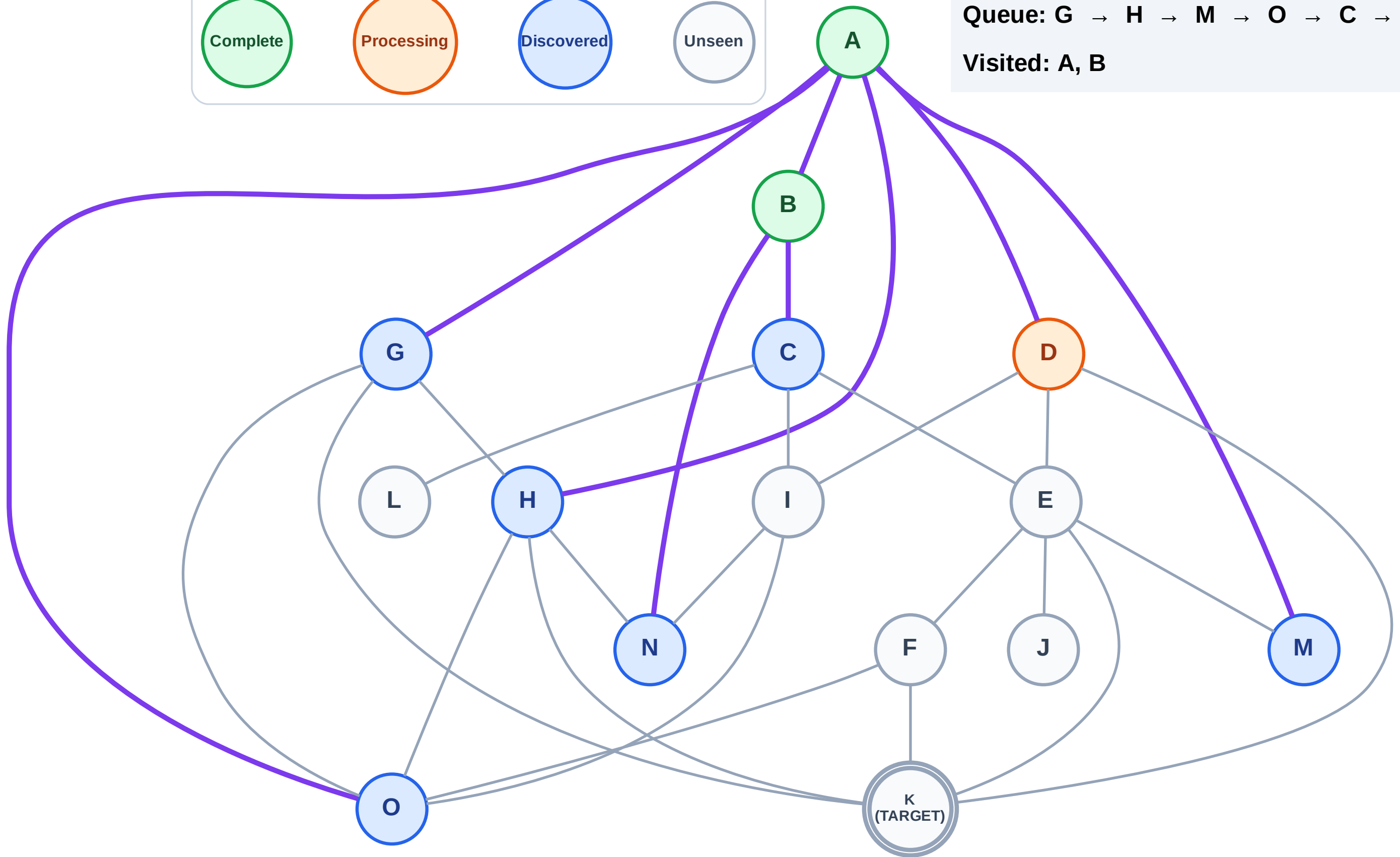
Step 6: Process node D

Legend



Queue: G → H → M → O → C → N

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

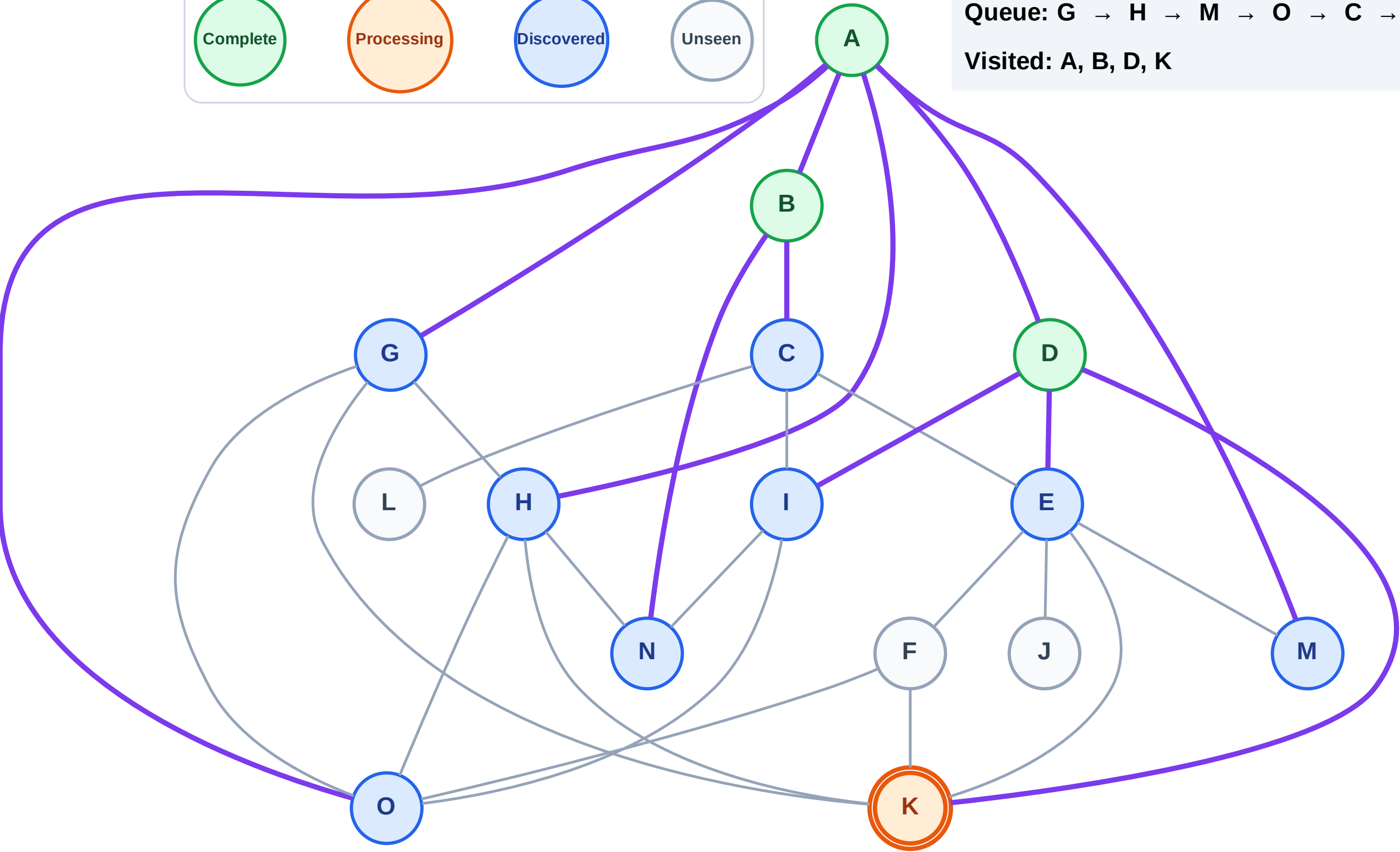
Processing

Discovered

Unseen

Queue: G → H → M → O → C → N → E → I

Visited: A, B, D, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

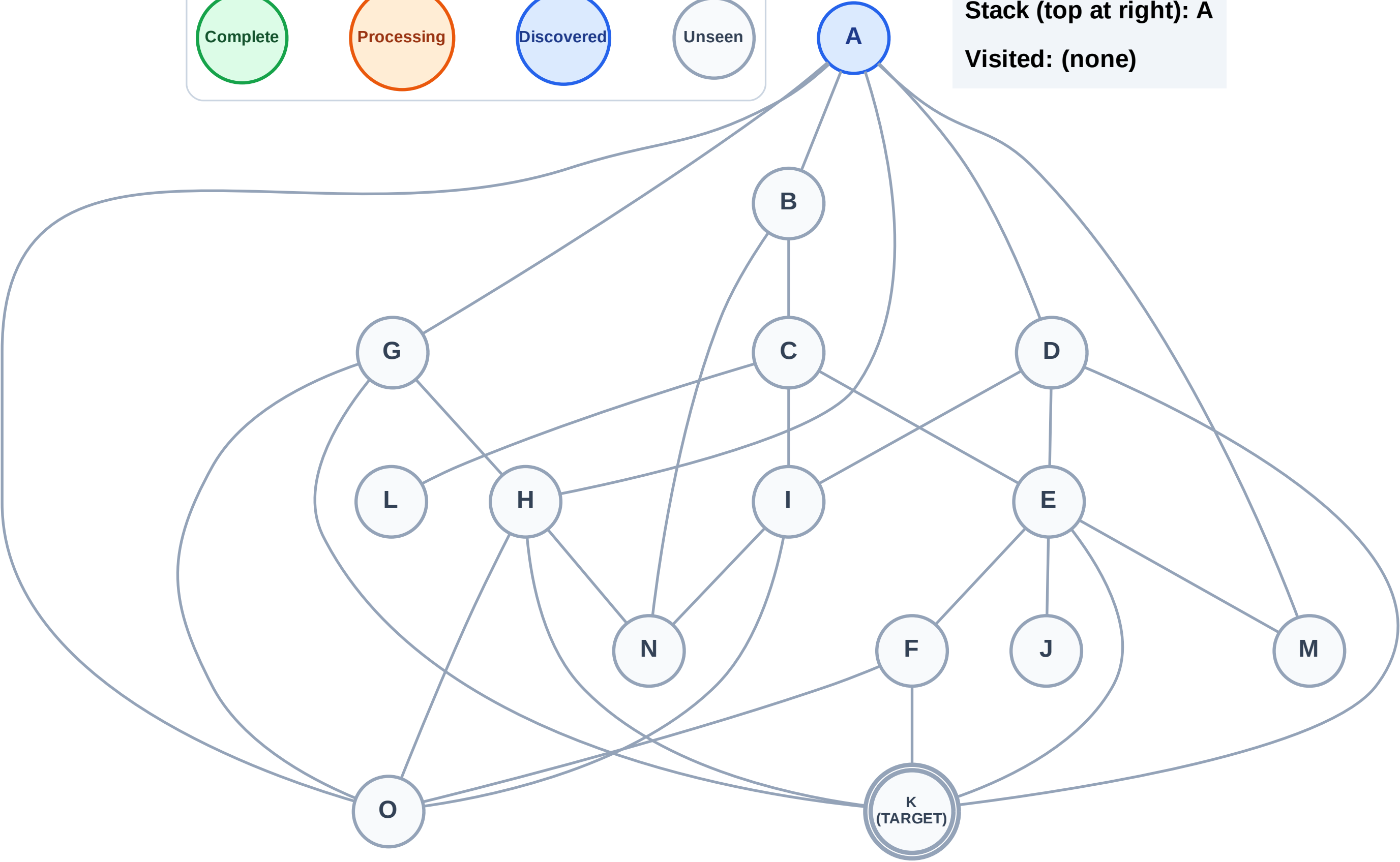
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

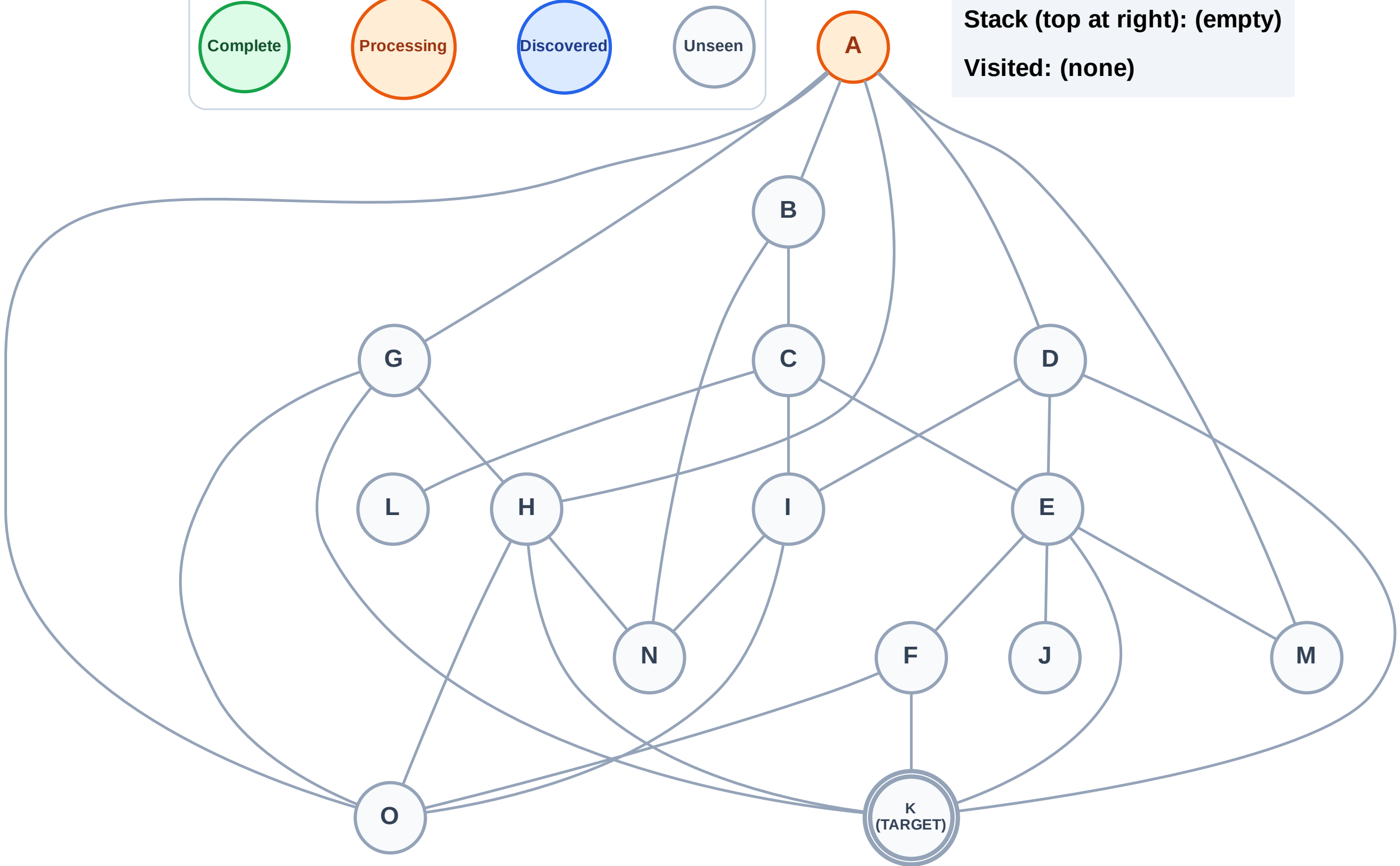
Complete

Processing

Discovered

Unseen

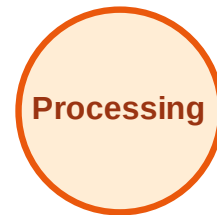
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

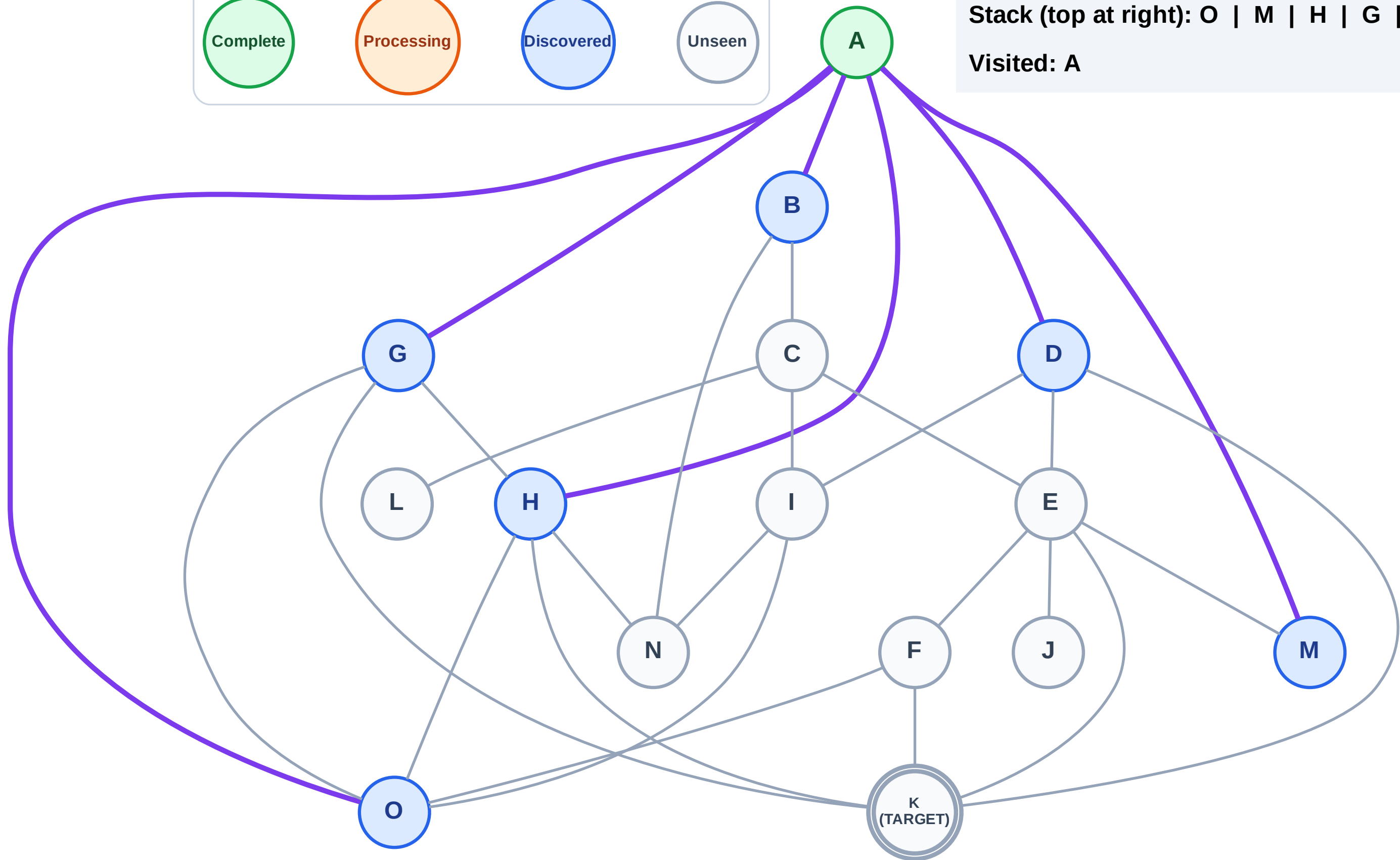
Step 3: Discover O, M, H, G, D, B from A

Legend



Stack (top at right): O | M | H | G | D | B

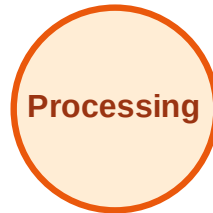
Visited: A



DFS Traversal

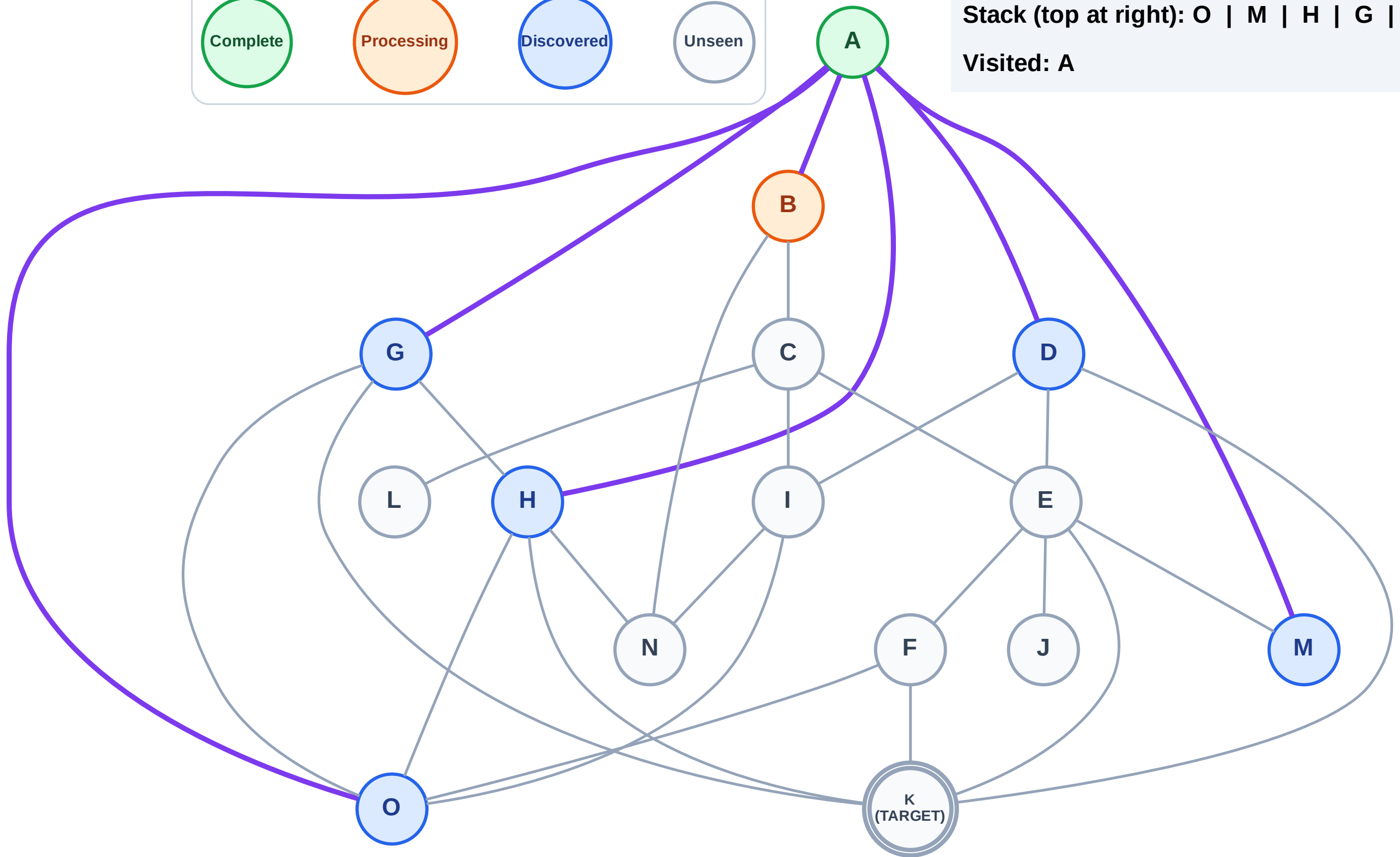
Step 4: Process node B

Legend



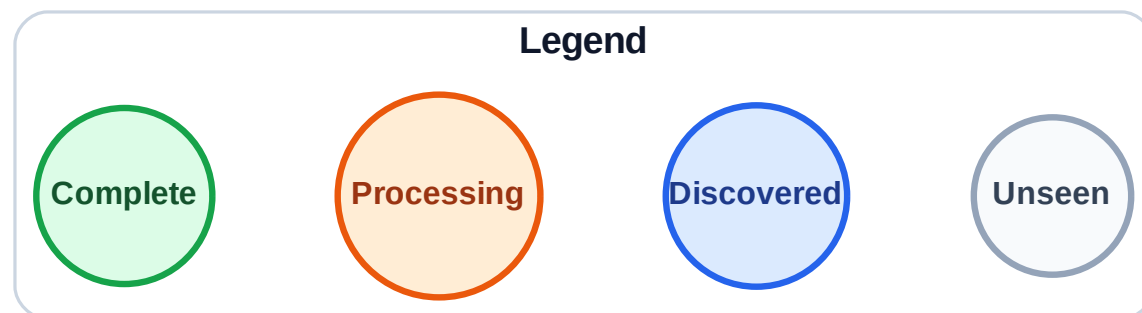
Stack (top at right): O | M | H | G | D

Visited: A



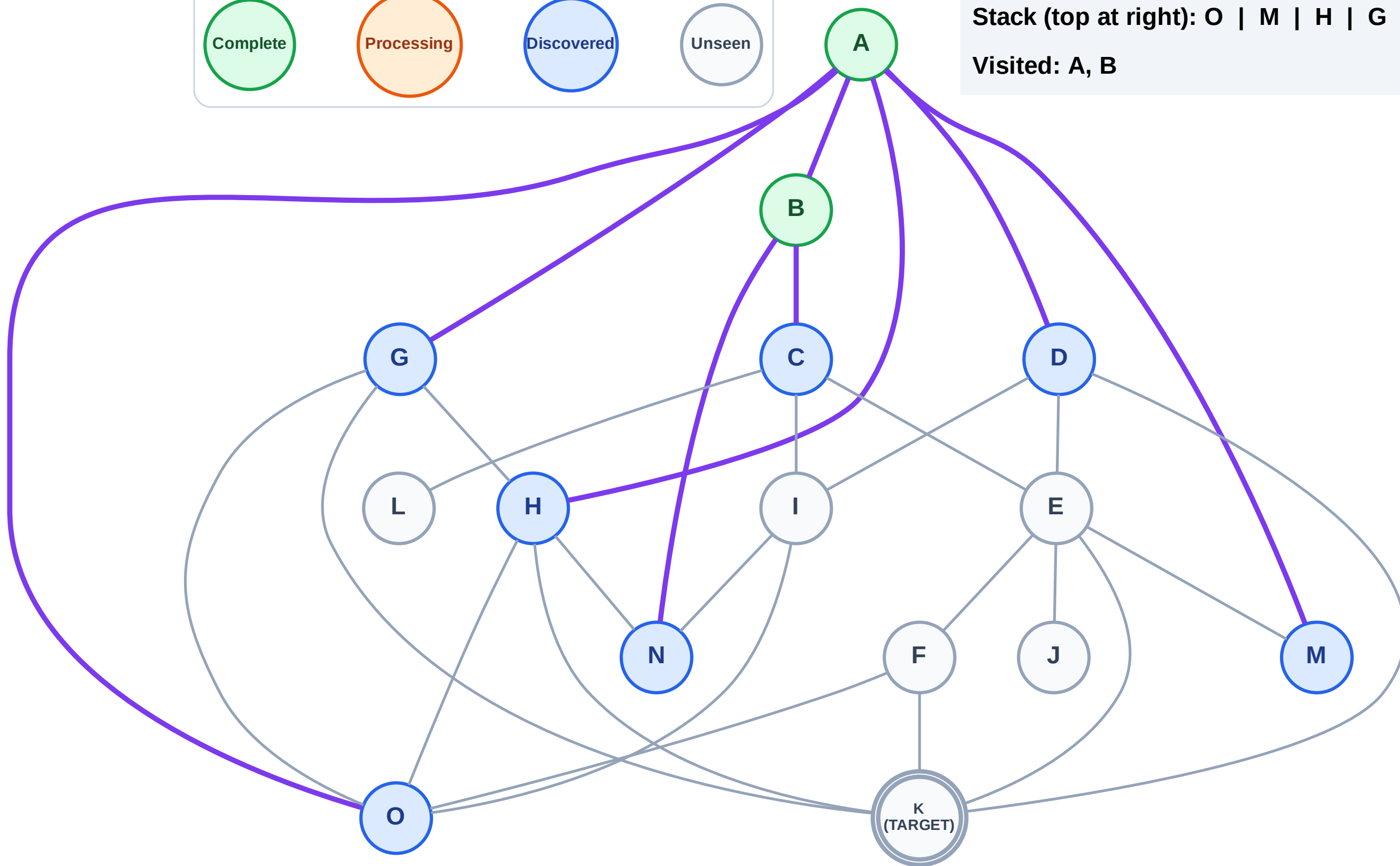
DFS Traversal

Step 5: Discover N, C from B



Stack (top at right): O | M | H | G | D | N | C

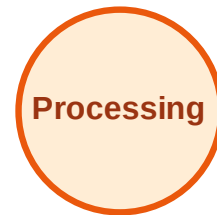
Visited: A, B



DFS Traversal

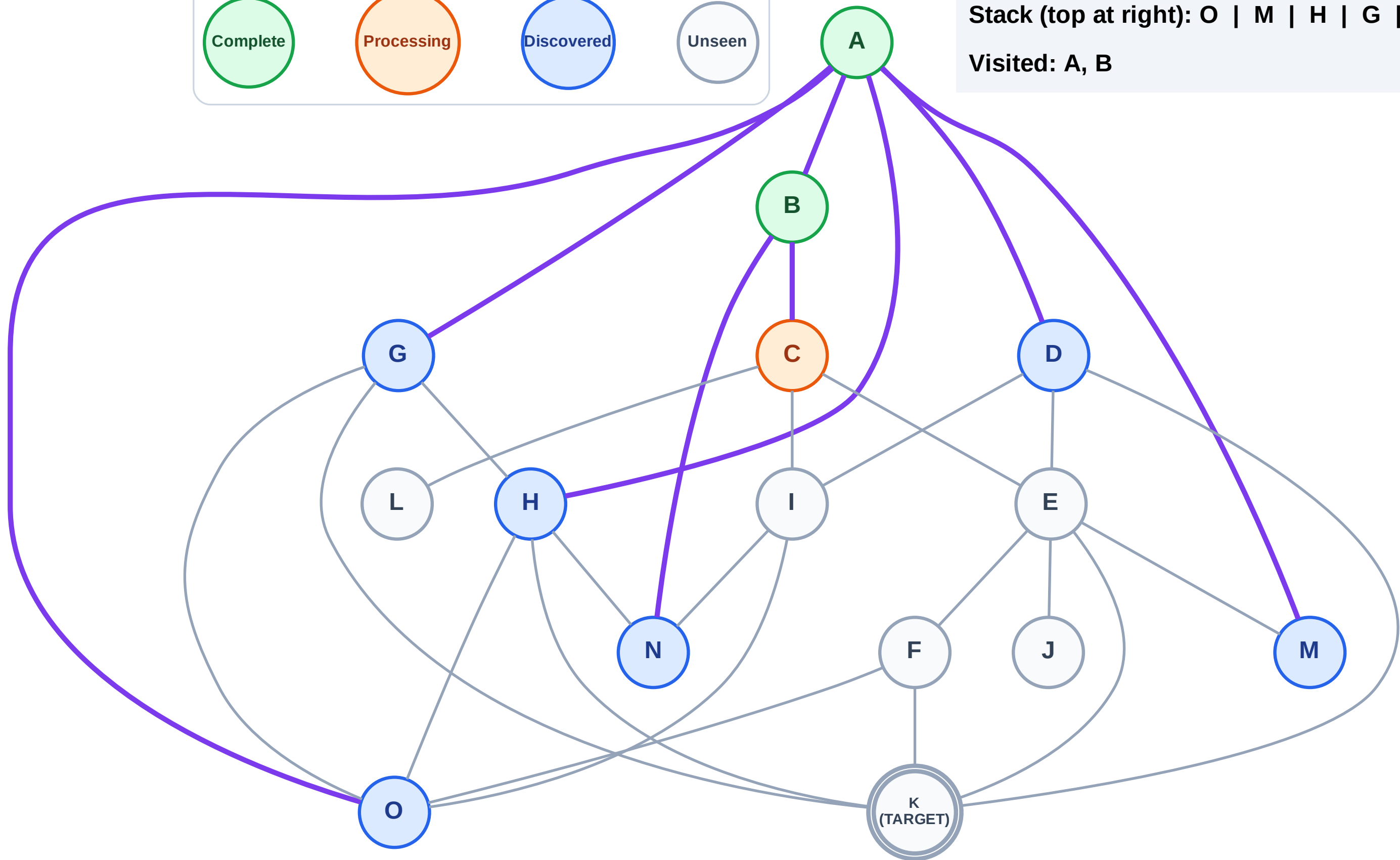
Step 6: Process node C

Legend



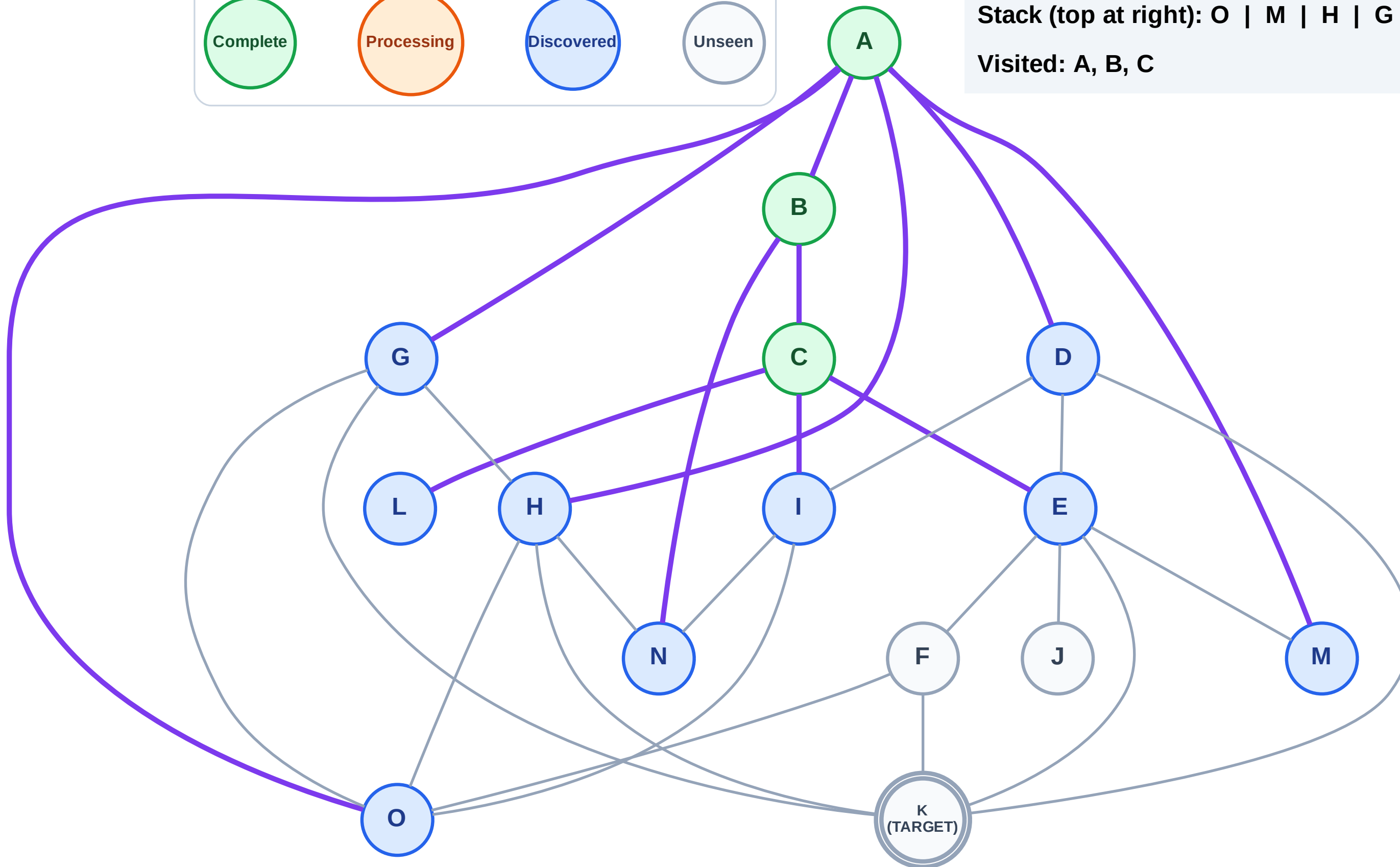
Stack (top at right): O | M | H | G | D | N

Visited: A, B



Step 7: Discover L, I, E from C

Legend



DFS Traversal

Step 8: Process node E

Legend

Complete

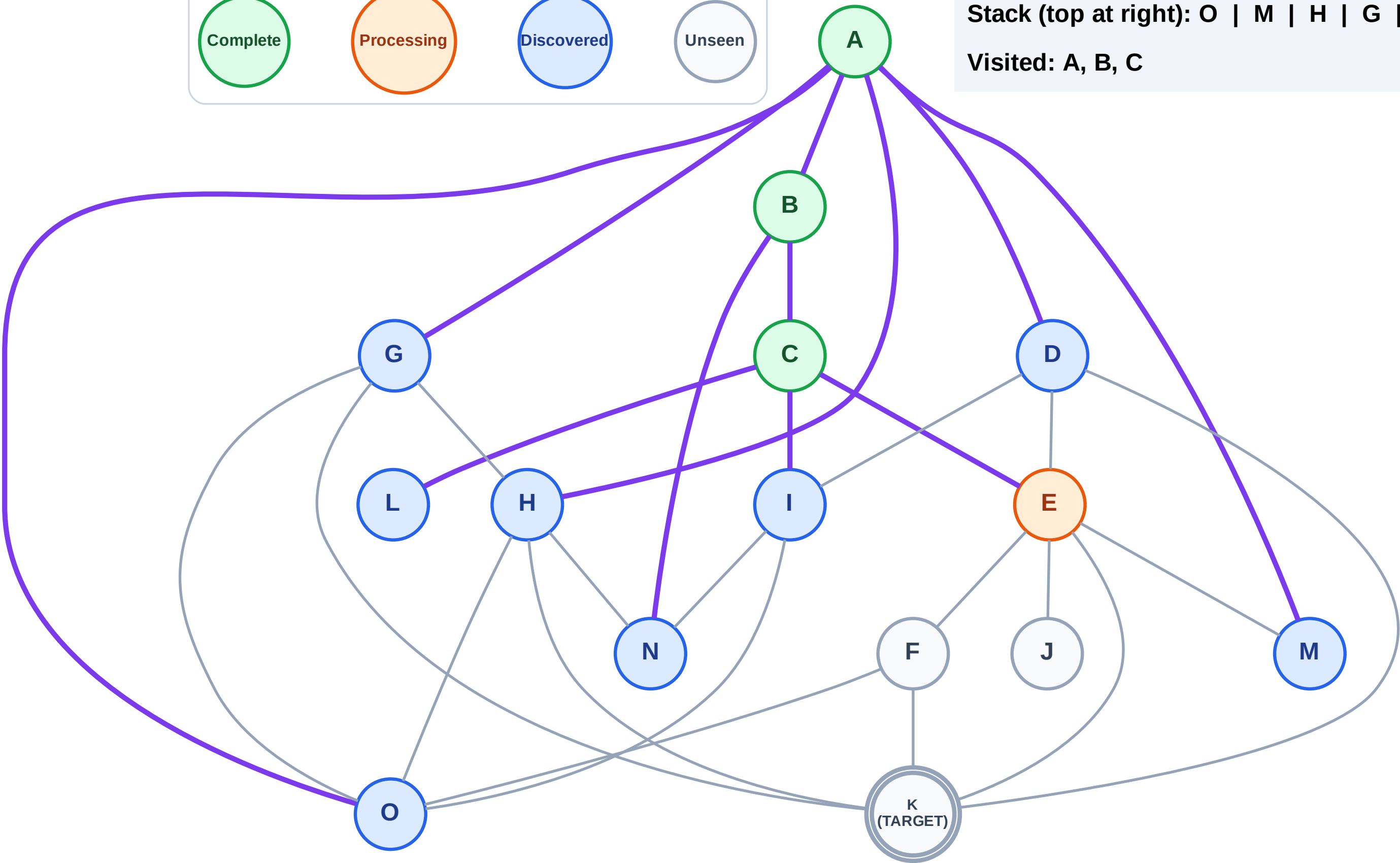
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I

Visited: A, B, C, E, K

